# GTPO-EMA: Theoretical Analysis

## Convergence to the Same Global Optimum as DAPO

An extension of GTPO with EMA-smoothed confidence weighting

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# Motivation: From GRPO to GTPO-EMA

**GRPO** assigns a uniform advantage to every token in a response:

$$\hat{A}_{i,t} = \hat{A}_i = \frac{r_i - \mathrm{mean}(\{r_k\})}{\mathrm{std}(\{r_k\})}$$

Every token in $o_i$ gets the same credit, regardless of uncertainty.

**GTPO** replaces this with an entropy-weighted per-token advantage.
**GTPO-EMA** further smooths the entropy signal via an EMA filter:

$$\mathrm{EMA}_{i,t} = \lambda \cdot \mathrm{EMA}_{i,t-1} + (1 - \lambda) \cdot C_{i,t}$$

## Key question

Does the EMA modification preserve the theoretical guarantees of GTPO, i.e., does GTPO-EMA still converge to the same optimum as GRPO/DAPO?

# Definitions: Confidence and EMA

## Definition 1 — Top-$k$ Confidence

$$C_{i,t} = -\operatorname*{mean}_{v \in \text{top-}k(\pi_\theta(\cdot \mid q, o_{i,<t}))} \log \pi_\theta(v \mid q, o_{i,<t})$$

$C_{i,t} \geq 0$; high value $\Leftrightarrow$ uncertain; low value $\Leftrightarrow$ confident.

## Definition 2 — EMA-smoothed Confidence ($\lambda \in (0,1)$)

$$\mathrm{EMA}_{i,0} = C_{i,0}, \qquad \mathrm{EMA}_{i,t} = \lambda \cdot \mathrm{EMA}_{i,t-1} + (1-\lambda) \cdot C_{i,t}$$

Expanding: $\mathrm{EMA}_{i,t} = (1-\lambda) \sum_{s=0}^{t} \lambda^s C_{i,t-s}$

Note: $\mathrm{EMA}_{i,t}$ is a *weighted average* of past confidence — recent steps have greater influence than distant ones.

# Definitions: Reward Shaping

Given successful/unsuccessful sets $O^+$, $O^-$ with active counts $d_t$, $h_t$:

**Successful sequence** $o_i \in O^+$ at position $t$:

$$\tilde{r}^+_{i,t} = \alpha_1 r_i + \alpha_2 \cdot \frac{\mathrm{EMA}_{i,t}}{\sum_{k \in O^+_t} \mathrm{EMA}_{k,t}} \cdot d_t$$

**Unsuccessful sequence** $o_j \in O^-$ at position $t$:

$$\tilde{r}^-_{j,t} = \alpha_1(-1) + \alpha_2 \cdot \frac{1/\mathrm{EMA}_{j,t}}{\sum_{k \in O^-_t} 1/\mathrm{EMA}_{k,t}} \cdot h_t \cdot (-1)$$

**Logic:**

- Uncertain success $\rightarrow$ larger bonus (valuable exploration)
- Confident failure $\rightarrow$ larger penalty (penalize overconfidence)

# Lemma 1 — Non-negativity of EMA

## Lemma

*For any sequence $o_i$ and timestep $t \geq 0$:* $\mathrm{EMA}_{i,t} \geq 0$.

**Proof** by induction on $t$:

**Base:** $\mathrm{EMA}_{i,0} = C_{i,0} \geq 0$ (by definition of $C_{i,t}$)

**Step:** Assuming $\mathrm{EMA}_{i,t-1} \geq 0$,

$$\mathrm{EMA}_{i,t} = \underbrace{\lambda}_{\geq 0} \cdot \underbrace{\mathrm{EMA}_{i,t-1}}_{\geq 0} + \underbrace{(1-\lambda)}_{\geq 0} \cdot \underbrace{C_{i,t}}_{\geq 0} \;\geq\; 0. \qquad \square$$

**Corollary:** $\sum_{k \in O_t^+} \mathrm{EMA}_{k,t} > 0$, so the normalization in the reward formula is well-defined.

# Lemma 2 — Deviation Formula

## Lemma

*For $o_i \in O_t^+$, let $\overline{\mathrm{EMA}}_t = \frac{1}{d_t}\sum_{k \in O_t^+} \mathrm{EMA}_{k,t}$. Then the shaped reward deviates from the flat GRPO reward by:*

$$\delta_{i,t} \;=\; \tilde{r}_{i,t}^{+} - r_i \;=\; \alpha_2\left(\frac{\mathrm{EMA}_{i,t}}{\overline{\mathrm{EMA}}_t} - 1\right)$$

**Proof:**

$$\begin{aligned}\delta_{i,t} = \tilde{r}_{i,t}^{+} - r_i &= \left[\alpha_1 + \alpha_2 \frac{\mathrm{EMA}_{i,t}}{\overline{\mathrm{EMA}}_t}\right] - 1 \\ &= -\alpha_2 + \alpha_2 \frac{\mathrm{EMA}_{i,t}}{\overline{\mathrm{EMA}}_t} = \alpha_2\left(\frac{\mathrm{EMA}_{i,t}}{\overline{\mathrm{EMA}}_t} - 1\right) \qquad \square\end{aligned}$$

*Interpretation:* $\delta_{i,t} = 0$ when token $i$ has exactly the group-average confidence. The deviation is zero-sum: $\sum_i \delta_{i,t} = 0$.

# Proposition 1 — Conservation of Reward Mass

## Proposition

*If $\alpha_1 + \alpha_2 = 1$, then at every timestep $t$:*

$$\sum_{i \in O_t^+} \tilde{r}_{i,t}^+ \;=\; \sum_{i \in O_t^+} r_i \;=\; d_t$$

**Proof:**

$$\sum_{i \in O_t^+} \tilde{r}_{i,t}^+ = \alpha_1 \underbrace{\sum_i r_i}_{d_t} + \alpha_2 d_t \underbrace{\sum_i \frac{\mathrm{EMA}_{i,t}}{\sum_k \mathrm{EMA}_{k,t}}}_{=1} = (\alpha_1 + \alpha_2)\, d_t = d_t \qquad \square$$

## Significance

The EMA shaping *redistributes* rewards among tokens but preserves the total reward mass. This holds for **any** non-negative weight function — EMA non-negativity (Lemma 1) is the only requirement.

# Assumption — EMA Consolidation

## Assumption (EMA Consolidation)

*As training progresses ($k \to \infty$), for any two successful sequences $o_{i_1}, o_{i_2} \in O_t^+$ (a.s.):*

$$\lim_{k\to\infty} \frac{\mathrm{EMA}_{i_1,t}}{\mathrm{EMA}_{i_2,t}} = 1 \qquad \Longleftrightarrow \qquad \lim_{k\to\infty} \frac{\mathrm{EMA}_{i,t}}{\overline{\mathrm{EMA}}_t} = 1$$

**Why reasonable:** As the policy converges, successful sequences develop a canonical reasoning pattern. Their confidence profiles $C_{i,t} \to \bar{C}$ become similar. Since $\mathrm{EMA}_{i,t} = (1-\lambda)\sum_s \lambda^s C_{i,t-s} \to \bar{C}$, the ratios $\mathrm{EMA}_{i,t}/\mathrm{EMA}_{j,t} \to 1$.

## Remark

*This is the EMA analogue of the Entropy Consolidation Condition used in the original GTPO analysis — it follows from it under the top-$k$ confidence approximation.*

# Proposition 2 — Asymptotic Consistency

## Proposition

*Under the EMA Consolidation assumption, with* $\|\nabla_\theta \log \pi_\theta\| \leq M$ *and bounded advantages:*

$$\lim_{k\to\infty} \|\Delta\mathcal{J}(\theta_k)\| = 0$$

*where* $\Delta\mathcal{J} = \nabla\mathcal{J}^{+}_{\textit{GTPO-EMA}} - \nabla\mathcal{J}_{\textit{GRPO}}$.

**Proof sketch:**

1. From Lemma 2 and EMA Consolidation: $\delta_{i,t} = \alpha_2\left(\frac{\mathrm{EMA}_{i,t}}{\overline{\mathrm{EMA}}_t} - 1\right) \to 0$
2. Shaped advantage converges: $\tilde{A}^{+}_{i,t} \to \hat{A}_{i,t}$ (Continuous Mapping Theorem, since $\mathrm{std}(R) > 0$)
3. Gradient bound: $\|\Delta\mathcal{J}\| \leq M \cdot \mathbb{E}\left[\frac{1}{G}\sum_i |\tilde{A}^{+}_{i,t} - \hat{A}_{i,t}|\right]$
4. Apply **Dominated Convergence** (bounded advantages): $\|\Delta\mathcal{J}\| \leq M \cdot \mathbb{E}[0] = 0 \qquad \square$

# Main Theorem

## Theorem — Same Global Optimum as DAPO

*Under the EMA Consolidation assumption and regularity conditions, GTPO-EMA shares the same global optimum $\theta^*$ as DAPO.*

**Proof:**

**Step 1.** At any critical point $\theta^*$ of $\mathcal{J}_{\text{GTPO-EMA}}$: $\nabla \mathcal{J}_{\text{GTPO-EMA}}(\theta^*) = 0$. By Proposition 2: $\nabla \mathcal{J}_{\text{GTPO-EMA}} \to \nabla \mathcal{J}_{\text{GRPO}}$, so $\nabla \mathcal{J}_{\text{GRPO}}(\theta^*) = 0$.

**Step 2.** DAPO differs from GRPO only by a constant reward shift ($0 \to -1$ for incorrect answers). This shifts the baseline but does not change the optimal direction. Both objectives share: $\theta^* = \arg\max \Pr[\text{correct} \mid q, \pi_\theta]$.

**Conclusion:**

$$\theta^*_{\text{GTPO-EMA}} = \theta^*_{\text{GRPO}} = \theta^*_{\text{DAPO}} \qquad \square$$

# Bonus Result — Variance Reduction

## Proposition

*Assuming $C_{i,t}$ have variance $\sigma_C^2$, as $t \to \infty$:*

$$\mathrm{Var}(\mathrm{EMA}_{i,t}) \;\to\; \frac{1-\lambda}{1+\lambda} \cdot \sigma_C^2$$

**Proof:** Since $\mathrm{EMA}_{i,t} = (1-\lambda)\sum_{s=0}^{t} \lambda^s C_{i,t-s}$ with i.i.d. $C_{i,s}$:

$$\mathrm{Var}(\mathrm{EMA}_{i,t}) = (1-\lambda)^2 \sum_{s=0}^{t} \lambda^{2s} \sigma_C^2 = \frac{(1-\lambda)^2}{1-\lambda^2}\big(1-\lambda^{2(t+1)}\big)\sigma_C^2 \xrightarrow{t\to\infty} \frac{1-\lambda}{1+\lambda}\sigma_C^2 \qquad \square$$

**Numerical example ($\lambda = 0.9$):**

| Weight signal | Variance | Reduction |
|---|---|---|
| Raw confidence $C_{i,t}$ | $\sigma_C^2$ | — |
| $\mathrm{EMA}_{i,t}$ ($\lambda = 0.9$) | $\approx 0.053\,\sigma_C^2$ | $\approx 19\times$ |

## Summary of Results

| Result | Statement |
|---|---|
| Lemma 1 | $\mathrm{EMA}_{i,t} \geq 0$ — normalization is always well-defined |
| Lemma 2 | $\delta_{i,t} = \alpha_2(\mathrm{EMA}_{i,t}/\overline{\mathrm{EMA}}_t - 1)$ — deviation is zero-sum |
| Prop. 1 | $\sum \tilde{r}^+_{i,t} = d_t$ — total reward mass is conserved |
| Prop. 2 | $\|\Delta\mathcal{J}(\theta_k)\| \to 0$ — gradient bias vanishes |
| **Theorem** | $\theta^*_{\text{GTPO-EMA}} = \theta^*_{\text{DAPO}}$ |
| Prop. 3 | $\mathrm{Var}(\mathrm{EMA}) \approx \sigma^2/19$ for $\lambda = 0.9$ |

### Take-away

GTPO-EMA does not change *where* the policy converges, but changes *how* — with a lower-variance, more informative gradient signal that rewards exploration on uncertain tokens and penalizes overconfident errors.

# Relationship Between Results

$$\underbrace{\mathrm{EMA}_{i,t} \geq 0}_{\text{Lemma 1}} \Rightarrow \underbrace{\sum \tilde{r}^{+}_{i,t} = d_t}_{\text{Prop. 1: conservation}}$$

$$\underbrace{\delta_{i,t} \to 0}_{\text{Lemma 2 + Assumption}} \Rightarrow \underbrace{\|\Delta \mathcal{J}\| \to 0}_{\text{Prop. 2: consistency}} \Rightarrow \underbrace{\theta^*_{\text{GTPO-EMA}} = \theta^*_{\text{DAPO}}}_{\textbf{Theorem}}$$